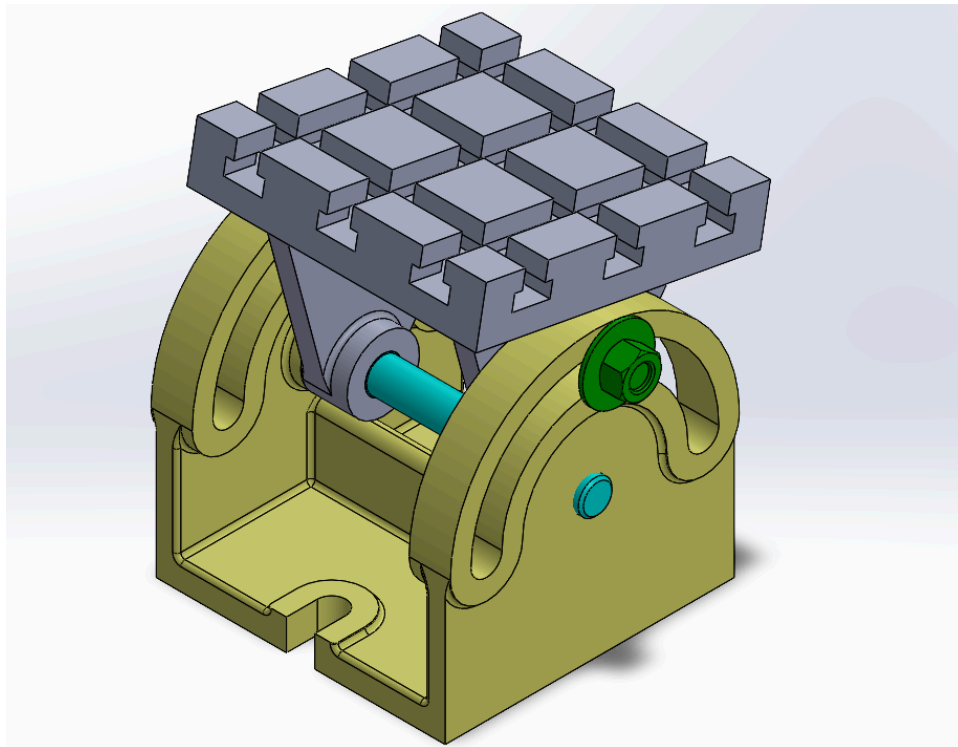


SWING TABLE ASSEMBLY

Design Development • SolidWorks Modeling • Assembly Documentation



Prepared by

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ME-430 — Machine Design

1. Project Overview

This project covers the design and 3D modeling of a swing table assembly, developed in SolidWorks from a reference engineering drawing. The assembly is built around a trunnion-pivot mechanism: a steel stud passes through bearing bores in a cast iron base and through matching arms on a cast iron table, allowing the table to rotate freely about a fixed horizontal axis while remaining rigidly supported.

The design exercise required three custom-modeled parts — the base, the table, and the trunnion stud — assembled with standard toolbox hardware, followed by a full assembly drawing, bill of materials, and detailed part drawings with complete dimensioning.

Design Intent

- **Pivoting motion:** The table swings freely about the trunnion stud, letting a workpiece be angled or indexed relative to the base.
- **Rigid mounting:** The base's T-slot channel and flat underside allow it to be clamped directly to a fixture or worktable.
- **Serviceable assembly:** All hardware — bolts, nuts, washers, and the set screw — is standard and readily sourced, keeping the design easy to assemble and maintain.

Materials

- **Base & Table:** Cast iron — chosen for rigidity, vibration damping, and dimensional stability under load.
- **Trunnion Stud:** Plain carbon steel — machined to a precision fit for smooth rotation in the base bores.
- **Fasteners:** AISI 304 stainless steel — hex bolt, hex nut, flat washer, and slotted set screw, sourced from the SolidWorks Toolbox.

2. Design Specifications

Key dimensions were taken directly from the reference assembly drawing and carried through into the SolidWorks models.

Dimension	Value
Overall footprint (base)	7" wide
Table plate	7-1/4" × 4-1/2"
Assembly height	5-5/8"
Pivot bore	3/4" core diameter
Table thickness / bore spacing	3-5/8" hole-to-hole

Bill of Materials

Materials were assigned to every component so the assembly drawing's BOM reports accurate part weights and material call-outs.

Item	Part	Material	Qty
1	Base	Ductile Iron (SN)	1
2	Table	Gray Cast Iron (SN)	1
3	Trunnion Stud	Plain Carbon Steel	1
4	Hex Nut, 5/8-11 UNC	AISI 304	2
5	Hex Bolt, 5/8-11 UNC × 2-1/2"	AISI 304	2
6	Flat Washer, 5/8" Regular	AISI 304	2
7	Set Screw, 3/8-16 UNC × 1/2", Slotted Oval Pt.	AISI 304	1

3. Assembled View

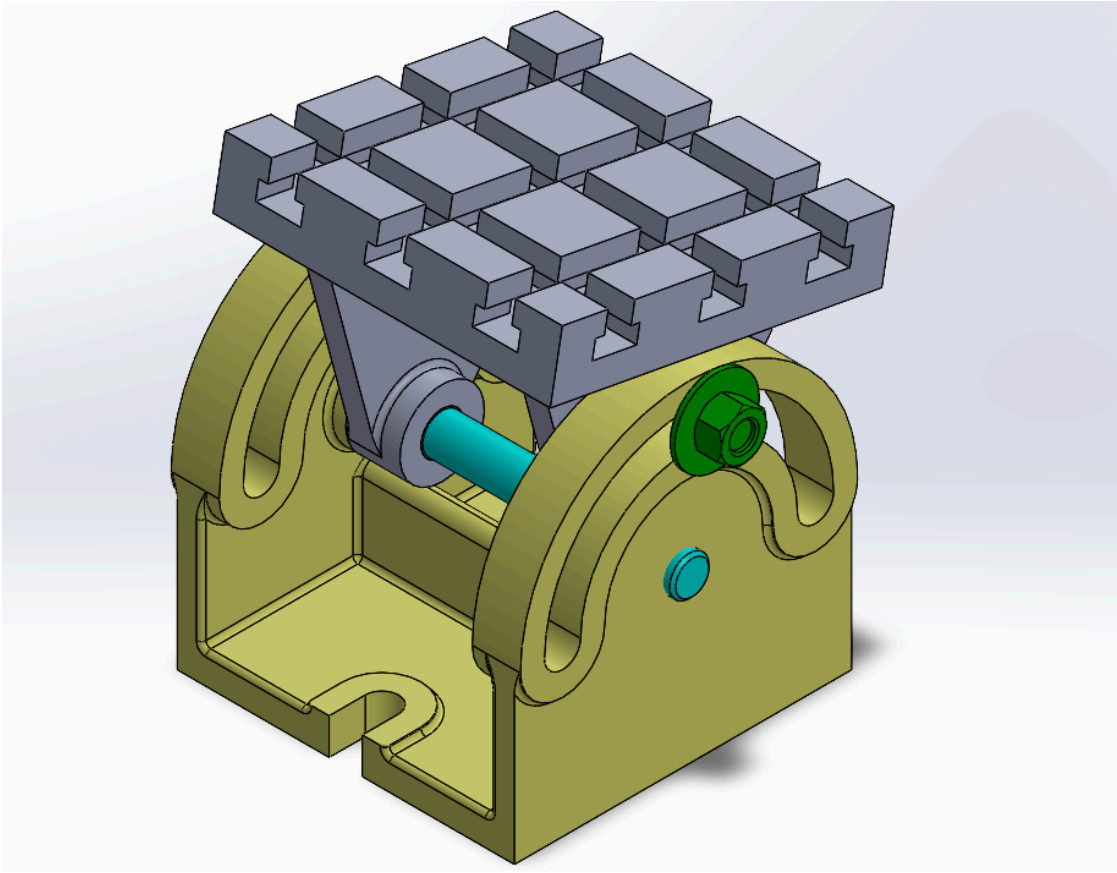


Fig. 1 — Swing table assembly, rendered in SolidWorks.

The trunnion stud is inserted through the base's bearing bores and the table's arms, then locked axially by the hex bolt and nut on one side and retained on the other by a slotted set screw bearing on the stud's flat. This lets the table pivot smoothly while eliminating end-play along the pivot axis.

4. Component Details

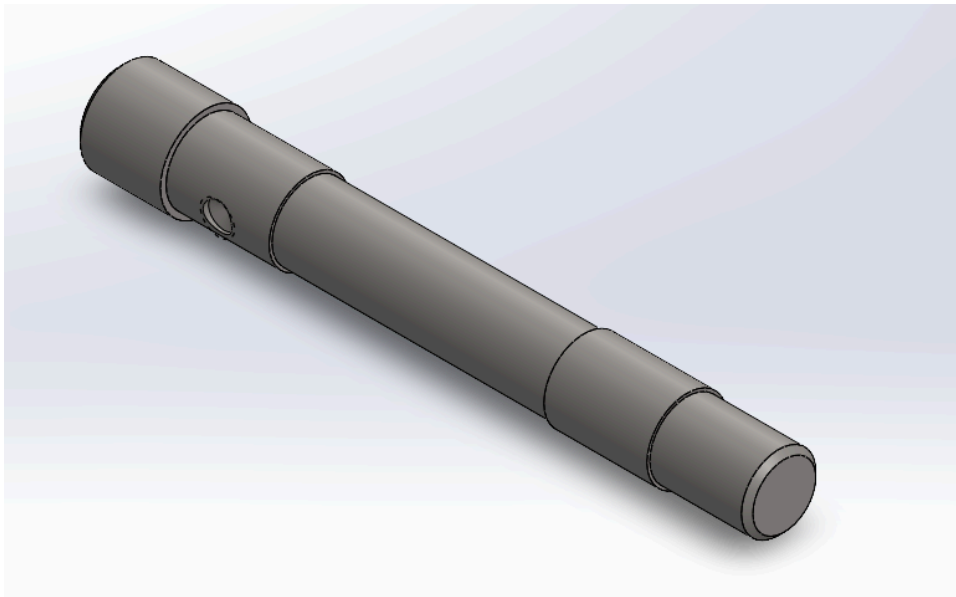
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Trunnion Stud



Precision steel pivot shaft — locates in the base bores and table arms, retained by the hex bolt/nut and set screw.